



Pietro Ferraiuolo

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Gender: Male **Date of birth**: 7 Oct 1996 **Nationality**: Italian

ABOUT MYSELF

I graduated in astronomy and astrophysics from 'La Sapienza - University of Rome,' driven by a fervent desire to become a scientific researcher. I am dedicated to a continuous self-improvement, with strong proficiency in technology and computer programming, particularly focused on data analysis. Beyond my academic pursuits, I'm an enthusiastic traveler who enjoys exploring new cultures. In my leisure time, I like to play video games, to read, and I love engaging in board games with my friends, as cherished hobbies.

CURRENT OCCUPATION

[15 Apr 2024 – Current] **Lead Software Developer for M4**

Research Fellowship: **Optical Qualification of the Test Tower for the ELT's M4 Adaptive Mirror and the DP Prototype.**

HONOURS AND AWARDS

[Feb 2025] **Grant di Ricerca Fondamentale Awarding institution**: INAF - Istituto Nazionale di Astrofisica

Awarded funding for the fundamental Astrophysics Call 2024. PI of the **BINCAT** project.

ONGOING RESEARCH

[Feb 2025 – Current] **BINCAT - Binary Catalog for AO Sytems Performace Evaluation**

Principal Investigator, INAF Fundamental Astrophysics Call 2024 (miniGrant Awardee)

Awarded funding for the development of a catalog of close binary stars (20–200 mas) as a tool for commissioning and optimizing adaptive optics (AO) systems on large telescopes. The project utilizes GAIA DR3 data to statistically identify multiplicity indicators and optimize the parameter space through PSF simulations. The catalog will support both current and next-generation AO instruments (e.g., MAVIS, MORFEO, ANDES).

[2024 – Current] **Optical Qualification of the Test Tower for the ELT's M4 Adaptive Mirror and the DP Prototype.**

Activities within the ESO E-ELT Project for the software development and calibration of the M4 Deformable Mirror (DM).

The main activities consist in:

- Software developing and debugging for the control of the Optical Test Tower designed for M4: motors for mechanical and optical alignment.
- Software developing and debugging for the DP (Demonstration Prototype) and M4 Deformable Mirrors (DMs).
- Data production and analysis for optical characterization and calibration of the M4 DM.

[May 2025 – Current] **ETIC Collaboration**

Collaboration within the OAA ETIC group for the development of CNN (*Convolutional Neural Networks*) for shape's command reconstruction for thermal-adaptive optics applications.

PRODUCTION

Publications

- N. Azzaroli, R. Briguglio, L. Oggioni, G. Pariani, C. Selmi, M. Xompero, P. Ferraiuolo, A. Riccardi; (in prep): "Subaperture Calibration of Deformable Mirrors for ELT-era DM size"

Software - GitHub Repositories

- **M4**
- **GRASP**: Globular cluster Astrometry and Photometry Software
- **OPTICALIB**: adaptive OPTics package for deformable mirrors CALibration
- **XuPy**: Cupy interface for GPU-accelerated Masked Arrays

Link: <https://github.com/pietroferraiuolo?tab=repositories>

EDUCATION AND TRAINING

[21 Jul 2025 – 1 Aug 2025]

GISM3 Doctoral School

UMR Astrophysique Instrumentation Modélisation (AIM) <https://ismgalaxies2025.sciencesconf.org/>

City: Banuyls-sur Mer | **Country**: France | **Field(s) of study**: The Interstellar Medium of Nearby Galaxies

Link: <https://drive.google.com/file/d/1RR80-jlmlbgTI5DI5EkzgE4iV-mH2pas/view>

[25 Feb 2025]

Course in Git, GitLab and CI/CD

CNR - Bologna & INAF <https://indico.ict.inaf.it/event/3067/>

[29 Sep 2024 – 5 Oct 2024]

AI-PHY Doctoral School

Università degli Studi di Milano - Bicocca <https://agenda.infn.it/event/40881/>

City: Monopoli (BA) | **Country**: Italy | **Field(s) of study**: Artificial Intelligence and modern PHYsics: a two-way connection | **Final grade**: Certificate of Attendance

Link: <https://drive.google.com/file/d/1nuGjUoGKD7ObFBU2cicM-DnC0NeYI-m6/view?usp=sharing>

[Jan 2021 – 23 Jan 2024]

Master's Degree in Astrophysics and Astronomy

Sapienza - University of Rome <https://www.uniroma1.it/it/pagina-strutturale/home>

Address: Piazzale Aldo Moro, 5, 00185, Roma, Italy |

Field(s) of study: Astronomy & Astrophysics ; Cosmology | **Final grade**: 110 | **Level in EQF**: EQF level 7 | **Thesis**: Dynamics of Globular Clusters: The Effective Potential through the Gaia Data Release 3

Link: <https://drive.google.com/file/d/114HtdUoHBJpqnzEidfCaNN9jyixtlnmW/view?usp=sharing>

WORK EXPERIENCE

2018-2020 Ohana Società Cooperativa Sociale Onlus | **2020-2022 Privately**

City: Pomezia (RM) | **Country**: Italy

[Sep 2018 – Sep 2022]

Learning Tutor

- Provided tutoring in scientific subjects, including mathematics and physics, to pre-teenagers and adolescents with SEN (Special Educational Needs), SLD (Specific Learning Disability), or other difficulties falling within the category of 'Learning Disorders'.
- Collaborated with a team, ensuring effective communication and management of educational support.
- Maintained regular contact with students' families, school staff, and specialized therapeutic professionals for planning and coordination.
- Contributed to team-building activities to foster a positive work environment among colleagues.

PROJECTS

[Mar 2023 – Dec 2023]

Dynamics of Globular Clusters: The Effective Potential Through the Gaia Data Release 3

Master's thesis research project, which explored the dynamics of collisional systems, in particular Globular Clusters. By analyzing GC phase-space data, their internal velocity distribution was used to extract the Effective Gravitational Potential within shells of the cluster and the cluster as a whole, providing a first observational counterpart to its theoretical derivation.

Link: https://drive.google.com/file/d/1Q-wPV1G5NUaBDUnpqOHGP_kyBJPQSKNM/view?usp=sharing

[May 2022 – Nov 2022]

Simulation of a point-like and an extended source of SZ signal and filtering optimization for MUSTANG2

Research experience provided by the master's degree, in which I had the opportunity to collaborate with Professor Elia Battistelli and his team in their project investigating the Cosmic Web through the Sunyaev-Zel'dovich effect measured in the hot gas in-between merging galaxy clusters. The project focused on simulating synthetic point-like and extended SZ sources and observing them with a CCD array similar to that of the GBT's MUSTANG-2 instrument.

Link: <https://drive.google.com/file/d/112wvZrSr0hAbSUMUNoCSjVqULpm8aPWA/view?usp=sharing>

CONFERENCES & SEMINARS

[19 Nov 2024]

AstroBigné INAF - Osservatorio Astrofisico di Arcetri

Short talk about my past and present research.

OUTREACH ACTIVITIES

[27 Sep 2024 – Current]

European Bright Night for Researchers

Interacted with the general public about astrophysical themes and the what is the job of researchers .

[Apr 2024 – Current]

Outreach Activities at the Observatory

Actively participating in the Observatory outreach activities, which include:

- Tour guide of the Observatory for schools and general public
- Technician and astronomer for observation nights, carried out with the MEADE telescopes.

SKILLS

Programming

Python | Git | R | Bash | C / C++ | Fortran90 | Linux | LaTeX

Language

English - C2 | Italian - Mother

